

Hyunji Amy Lee

hyunji.amy.lee@kaist.ac.kr | <https://amy-hyunji.github.io/>

RESEARCH INTEREST

Natural Language Processing and Multimodal AI: knowledge-augmented language models, retrieval and memory systems, multi-agent AI, knowledge grounding, reasoning, lifelong learning, and multimodal understanding.

My research goal is to build reliable, adaptable, and generalizable AI systems capable of solving complex real-world tasks. To achieve this, I study how language and multimodal models can effectively leverage and reason over diverse knowledge modules, including retrieved documents, long-term memories, specialized agents, external tools, multimodal knowledge repositories, and the model's own parametric knowledge. My research focuses on developing methods that align language models with heterogeneous knowledge sources, enabling models to identify, select, integrate, and ground information from the most appropriate knowledge modules for a given task. I investigate how multiple knowledge modules can be incorporated into a unified reasoning framework, improving performance in long-horizon, knowledge-intensive, and continually evolving environments. More broadly, I am interested in building trustworthy and controllable AI systems that can combine parametric and non-parametric knowledge, perform grounded reasoning, and generalize across text, code, and multimodal domains.

EDUCATION

KAIST AI

Ph.D. in AI Grad School. Advisor: [Minjoon Seo](#) Mar. 2023 – Aug. 2025
Committee: Eunsol Choi, Edward Choi, Arman Cohan, James Thorne
Thesis: Aligning Language Models with Knowledge Modules

KAIST AI

M.S. in AI Grad School. Advisor: [Minjoon Seo](#) Mar. 2021 – Feb. 2023

KAIST

B.S. in Electrical Engineering & Computer Science (Double Major) Mar. 2016 – Feb. 2021

EXPERIENCE

University of North Carolina, Chapel Hill

Postdoctoral Research Associate (Advisor: Mohit Bansal) North Carolina, US Aug 2025~

Microsoft Research

Research Intern (Advisors: Lucas Caccia, Marc-Alexandre Cote, Eric Xingdi Yuan, Alessandro Sordoni) Montreal, Canada May 2025~ Aug 2025 [C16]

Tencent AI Lab

Research Intern, Tencent AI Lab (Advisors: Wenhao Yu, Hongming Zhang, Kaixin Ma) Seattle, US Sep 2024 ~ Apr 2025 [W2]

Adobe Research

Research Intern, Vision Language Lab (Advisors: David Seunghyun Yoon, Trung Bui, Franck Dernoncourt) San Jose, US May 2024 ~ Aug 2024 [C13]

Allen Institute for AI (AI2)

Research Intern, Semantic Scholar Team (Advisors: Kyle Lo, Luca Soldani, Arman Cohan) Seattle, US May. 2023 - Dec. 2023 [P2, C10]

MindLogic

Seoul, Republic of Korea

Research Scientist Intern (Advisor: YongWoo Kim)
Developing a real-time korean QA Chatbot model

Dec. 2020– Feb. 2020

Naver

Research Scientist Intern (Mentor: Sohee Yang, Wonseok Hwang, Minjoon Seo)
Developing a cost-effective parser model for semi-structured document images

Seongnam, Republic of Korea

Mar. 2020 – Dec. 2020
[C1]

SKT AI

AI Fellowship (Mentor: Eric Davis, Jaehoon Kim)
Building Efficient Koran Multi Sentiment Analysis Model ([page](#))

Seoul, Republic of Korea

June. 2019 – June. 2020

MindLogic

Research Scientist Intern (Mentor: YongWoo Kim)
Developing Virtual Try-On Model ([page](#))

Seoul, Republic of Korea

Dec. 2019 – Feb. 2020

Naver

Research Scientist Intern, Developing Auto Colorization Model for Webtoon
Mentor: Gayoung Lee

Jeongja, Republic of Korea

June. 2018 – Dec. 2018

PUBLICATIONS

Conference and Journal Papers

[C19] Multimodal Fact-Level Attribution for Verifiable Reasoning

David Wan*, Han Wang*, Ziyang Wang, Elias Stengel-Eskin, [Hyunji Lee](#), Mohit Bansal
Accepted to ICML 2026

[C18] Generalized correctness models: Learning calibrated and model-agnostic correctness predictors from historical patterns

Hanqi Xiao, Vaidehi Patil, [Hyunji Lee](#), Elias Stengel-Eskin, Mohit Bansal
Accepted to ICML 2026

[C17] Can Large Language Models Keep Up? Benchmarking Online Adaptation to Continual Knowledge Streams

Jiyeon Kim*, [Hyunji Lee](#)*, Dylan Zhou*, Sue Hyun Park, Seunghyun Yoon, Trung Bui, Franck Dernoncourt, Sungmin Cha, Minjoon Seo
Accepted to ACL 2026

[C16] Gistify! Codebase-Level Understanding via Runtime Execution

[Hyunji Lee](#)*, Minseon Kim, Chinmay Singh, Matheus Pereira, Atharv Sonwane, Isadora White, Elias Stengel-Eskin, Mohit Bansal, Zhengyan Shi, Alessandro Sordoni, Marc-Alexandre Cote, Xingdi Yuan, Lucas Caccia*
Accepted to ICLR 2026

[C14] Instruction Tuning with and without Context: Behavioral Shifts and Downstream Impact

[Hyunji Lee](#), Seunghyun Yoon, Yunjae Won, Hanseok Oh, Geewook Kim, Trung Bui, Franck Dernoncourt, Elias Stengel-Eskin, Mohit Bansal, Minjoon Seo
Accepted to EACL 2026 (Long) [Oral](#)

[C13] Enhancing Language Models to Efficiently Generate All Relevant Answers from Complex, Interrelated Contexts

[Hyunji Lee](#), Franck Dernoncourt, Trung Bui, Seunghyun Yoon
Accepted to NACCL 2025 (Long)

- [C12] **How Does Vision-Language Adaptation Impact the Safety of Vision Language Models?**
Seongyun Lee*, Geewook Kim*, Jiyeon Kim*, Hyunji Lee, Hoyeon Chang, Sue Hyun Park, Minjoon Seo
Accepted to ICLR 2025
- [C11] **Knowledge Entropy Decay during Language Model Pretraining Hinders New Knowledge Acquisition**
Jiyeon Kim*, Hyunji Lee*, Hyowon Cho, Joel Jang, Hyeonbin Hwang, Seungpil Won, Youbin Ahn, Dohaeng Lee, Minjoon Seo
Accepted to ICLR 2025 [Oral](#)
AAAI KnowFM Workshop 2025 [Oral \(Best Paper Awards\)](#)
- [C10] **RouterRetriever: Exploring the Benefits of Routing over Multiple Expert Embedding Models**
Hyunji Lee, Luca Soldani, Arman Cohan, Minjoon Seo, Kyle Lo
Accepted to AAAI 2025 Main Technical Track
- [C9] **Exploring the practicality of generative retrieval on dynamic corpora**
Chaeun Kim*, Soyoung Yoon*, Hyunji Lee, Joel Jang, Minjoon Seo
Accepted to EMNLP 2024 (Long)
- [C8] **SemiParametric Token-Sequence Co-Supervision**
Hyunji Lee*, Doyoung Kim*, Jihoon Jun, Sejune Joo, Joel Jang, Kyung-Woon On, Minjoon Seo
Accepted to ACL 2024 (Long)
AAAI KnowFM Workshop 2025
- [C7] **How Well Do Large Language Models Truly Ground?**
Hyunji Lee*, Sejune Joo*, Chaeun Kim, Joel Jang, Kyung-Woon On, Minjoon Seo
Accepted to NACCL 2024 (Long) [Oral](#)
ACL KnowledgeNLP Workshop 2024 [Oral \(Best Student Paper\)](#)
- [C6] **KTRL+F: Knowledge-Augmented In-Document Search**
Hanseok Oh*, Haebin Shin*, Miyoung Ko, Hyunji Lee, Minjoon Seo
Accepted to NACCL 2024 (Long)
ACL KnowledgeNLP Workshop 2024, ACL KnowledgeLM Workshop 2024
- [C5] **Improving Probability-based Prompt Selection Through Unified Evaluation and Analysis**
Sohee Yang, Jonghyeon Kim, Joel Jang, Seonghyeon Ye, Hyunji Lee, Minjoon Seo
Accepted to TACL 2024
- [C4] **Nonparametric Decoding for Generative Retrieval**
Hyunji Lee, Jaeyoung Kim, Hoyeon Chang, Hanseok Oh, Sohee Yang, Vlad Karpukhin, Yi Lu, Minjoon Seo
Accepted to Findings of ACL 2023 (Long)
- [C3] **Local 3D Editing via 3D Distillation of CLIP Knowledge**
Junha Hyung, Sungwon Hwang, Daejin Kim, Hyunji Lee, Jaegul Choo
Accepted to CVPR 2023
- [C2] **Generative Multi-hop Retrieval**
Hyunji Lee, Sohee Yang, Hanseok Oh, Minjoon Seo
Accepted to EMNLP 2022 (Long) [Oral](#), AKBC 2022
[Finalist](#) Qualcomm Innovation Fellowship 2023
- [C1] **Cost-Effective End-to-End Information Extraction for Semi-Structured Document Images**
Wonseok Hwang, Hyunji Lee, Jinyeong Yim, Geewook Kim, Minjoon seo
Accepted to EMNLP 2021 (Short)

Posters and Workshop Papers

[W1] InstructIR: A Benchmark for Instruction Following of Information Retrieval Models

Hanseok Oh, [Hyunji Lee](#), Seonghyeon Ye, Haebin Shin, Hansol Jang, Changwook Jun, Minjoon Seo
ACL KnowledgeNLP Workshop 2024 [Oral](#)

[W2] Understanding and Enhancing Mamba-Transformer Hybrids for Memory Recall and Language Modeling

[Hyunji Lee](#), Wenhao Yu, Hongming Zhang, Kaixin Ma, Jiyeon Kim, Dong Yu, Minjoon Seo
EMNLP BabyLM Workshop 2025

Preprints / Under Review

[P10] AVSD: Adaptive-View Self-Distillation by Balancing Consensus and Teacher-Specific Privileged Signals

Duy Nguyen, Hanqi Xiao, Archiki Prasad, Zaid Khan, Anirban Das, Austin Zhang, Sambit Sahu, [Hyunji Lee](#), Elias Stengel-Eskin, Mohit Bansal
Under Review

[P9] MINTEVAL: Evaluating Memory under Multi-Target Interference in Long-Horizon Agent Systems

[Hyunji Lee](#)*, Justin Chih-Yao Chen*, Joykirat Singh, Zaid Khan, Elias Stengel-Eskin, Mohit Bansal
Under Review

[P8] EgoMemReason: A Memory-Driven Reasoning Benchmark for Long-Horizon Egocentric Video Understanding

Ziyang Wang, Yue Zhang, Shoubin Yu, Ce Zhang, Zengqi Zhao, Jaehong Yoon, [Hyunji Lee](#), Gedas Bertasius, Mohit Bansal
Under Review

[P7] Agent-BRACE: Decoupling Beliefs from Actions in Long-Horizon Tasks via Verbalized State Uncertainty

Joykirat Singh, Zaid Khan, Archiki Prasad, Justin Chih-Yao Chen, Akshay Nambi, [Hyunji Lee](#), Elias Stengel-Eskin, Mohit Bansal
Under Review

[P6] Balancing Faithfulness and Performance in Reasoning via Multi-Listener Soft Execution

Nithin Sivakumaran, Shoubin Yu, [Hyunji Lee](#), Yue Zhang, Ali Payani, Mohit Bansal, Elias Stengel-Eskin
Under Review

[P5] Conflict-Resolving and Sharpness-Aware Minimization for Generalized Knowledge Editing with Multiple Updates

Duy Nguyen, Hanqi Xiao, Archiki Prasad, Elias Stengel-Eskin, [Hyunji Lee](#), Mohit Bansal
Under Review

[P4] BugPilot: Complex Bug Generation for Efficient Learning of SWE Skills

Atharv Sonwane*, Isadora White*, [Hyunji Lee](#), Matheus Pereira, Lucas Caccia, Minseon Kim, Zhengyan Shi, Chinmay Singh, Alessandro Sordani, Marc-Alexandre Cote, Xingdi Yuan
Under Review

[P3] Differential Information: An Information-Theoretic Perspective on Preference Optimization

Yunjae Won, [Hyunji Lee](#), Hyeonbin Hwang, Minjoon Seo
Under Review

[P2] Back to Basics: A Simple Recipe for Improving Out-of-Domain Retrieval in Dense Encoders

[Hyunji Lee](#), Luca Soldani, Arman Cohan, Minjoon Seo, Kyle Lo
Preprint arxiv

[P1] Zero-Shot Dense Video Captioning by Jointly Optimizing Text and Moment
Yongrae Jo, Seongyun Lee, Aiden SJ Lee, Hyunji Lee, Hanseok Oh, Minjoon Seo
Preprint arxiv

HONORS, AWARDS, AND SCHOLARSHIPS

MLCommons Rising Stars <i>MLCommons Rising Stars 2026</i>	2026
Best Paper Awards <i>AAAI KnowFM Workshop 2024</i>	2025
Best Student Paper <i>ACL KnowledgeNLP Workshop 2024</i>	2024
GCP Credit Awards <i>Gemini Academic Program</i>	2024
Qualcomm Innovation Fellowship <i>Qualcomm, Finalist</i>	2023
KAIST Graduation Leadership Awards <i>KAIST</i>	2021
Qualcomm-KAIST Multimodal Emotion Recognition Kaggle <i>Qualcomm, 5th Prize</i>	2020
SKT AI Fellowship <i>SKT AI</i>	2019
DKU (Duke & Duke-Kunshan Univ.) program as KAIST representative <i>Duke & Duke-Kunshan Univ.</i>	2018
KAIST Global Leadership Awards <i>KAIST</i>	2018
KAIST Leadership & Volunteering Awards <i>KAIST</i>	2017